Ionising Power & Deflection

Exam code: 0625 & 0972

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Written by: Ashika
Reviewed by: Caroline Carroll
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Ionising effect of radiation

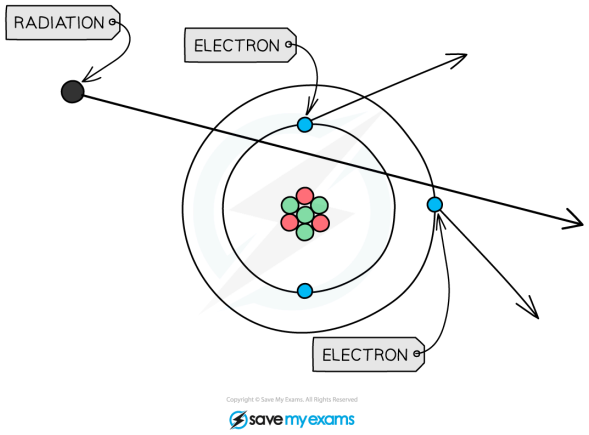
Extended tier only

- The relative ionising effects of nuclear radiation depend upon their:
  - **kinetic energy**
  - **electric charge**

Ionisation

- Ionisation is when an atom becomes negatively or positively charged by gaining or losing electrons
- Nuclear radiation can **ionise** the atoms that it hits
  - This is mostly done by removing an electron so the atom loses a negative charge and is left with an overall **positive** charge

Nuclear radiation ionising an atom



When radiation passes close to atoms it can knock out electrons, ionising the atom

Effects of kinetic energy and charge on ionising power

- The **greater** the **charge** of the radiation, the **more** ionising it is

Help me understand

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Ask questions and get help with exactly what you're revising.
Got it

ionising
    - This means gamma radiation is the least ionising as it has a charge of 0 (no charge)

- The **higher** the **kinetic energy** of the radiation, the **more** ionising it is
    - This means the alpha particle is still the most ionising because it has the greatest mass
    - However, a beta particle is very light (it is an electron) but travels at high speeds, therefore, it has a lot of kinetic energy and is still moderately ionising
    - Gamma radiation has virtually no mass so it is weakly ionising

Examiner Tips and Tricks

Remembering the properties of alpha, beta and gamma radiation helps to deduce how much ionising power they have. E.g. An alpha particle is a helium nucleus which contains two protons and two neutrons. It therefore has a charge of +2 since each proton has a charge of +1 and a neutron has no charge.

Kinetic energy is defined by the equation $\frac{1}{2}mv^2$ therefore it depends on the mass m of the particle and its velocity v.

Deflection in electric & magnetic fields

Extended tier only

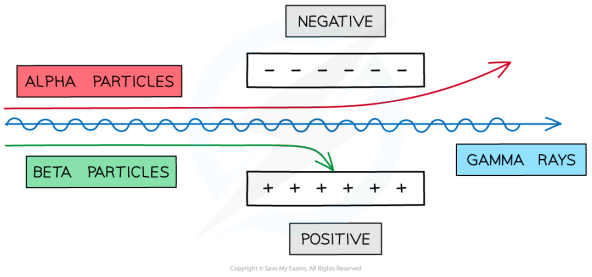
- α-particles, β-particles and γ-radiation are deflected differently in electric and magnetic fields
- A particle is deflected in an electric field if it has **charge**
- A particle is deflected in a magnetic field if it has **charge** and is **moving** perpendicular to it
    - Therefore, since gamma (γ) particles have no charge, they are **not** deflected by either electric or magnetic fields
    - Only alpha (α) and beta (β) particles are deflected

Electric fields

- Alpha particles have a **charge of +2** (the charge of a helium nucleus)
- Beta particles have a **charge of −1** (the charge of an electron)
- Therefore, in an electric field created between negatively and positively charged plates
    - Alpha particles are deflected towards the **negative** plate

through between the plates

Deflection in electric fields



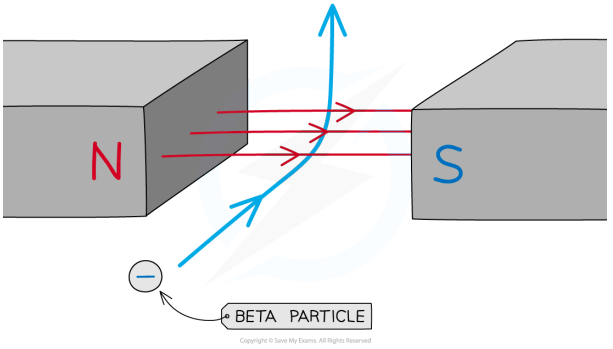
Alpha and Beta particles can be deflected by electric fields

- Alpha particles are **heavier** than beta particles
  - Therefore, **beta** particles are deflected **more** in the electric field

Magnetic fields

- Similarly, alpha and beta particles are deflected by magnetic fields whilst they are moving
- They are deflected in **opposite** directions due to their opposite charges

Deflection of a beta particle in a magnetic field



Alpha and beta particles can also be deflected by magnetic fields

Examiner Tips and Tricks

It is important to note that because of their opposite charges, alpha and beta particles will deflect in **opposite** directions. You need to know in which direction alpha and beta particles are deflected in an electric field and a magnetic field.

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